

Can You Put a Number on Learning

(You can try)

Rishabh Singh

Introduction

Using a survey from students in CMSC 320, along with their results from their first midterm in 320 two studies were conducted: one used machine learning algorithms to predict the letter grade students received, whether they passed or failed, or predict their exact grade; the latter half of the study was to determine what the largest factor in predicting a student's grade was in an effort to better prepare for future exams.

In predicting student scores in categorical forms, Decision Trees, K nearest neighbor (KNN), and Support Vector Machine (SVM) with reasonable accuracies for passing/failing predictions and lower accuracies for classifying letter grades. For predicting specific grades, a linear regression model was followed with L1, L2, and Elasticnet models, all of which did poorly on predicting actual scores.

Lastly, random forest trees were used in determining which factors had the largest impact on predicting a student's grade. The findings showcased that the order of which students submitted the survey had a large impact. Whether or not that is a meaningful predictor is not fully certain.

Background

This survey was taken after the exam concluded but before the results were released. It consisted of some questions directly related to preparation for the test, such as "How many hours did you study?" or "Did you do the readings?" as well as questions less related to the test like "How many hours a day on average do you spend on sites with infinite scroll?". All of these being answered in a multiple-choice fashion. The survey also automatically recorded time and date for each response as well as the number response it was. This gives us a diverse set of information on which we can train our models and learn which factors impact results the most. One important note is that results on the exam were lower than normal, with a median of 64% and standard deviation of 15% which may have made this data set less ideal for comparing to normal exams.

Data Cleaning

All questions were multiple choice, so no cleaning is needed regarding mistyped results. Some respondents were missing data for the time of class questions, but those rows were just removed without having much affect on the overall dataset. The other piece of cleaning was removing data for students who answered yes to the “Did you put random answers question?”, as they shouldn’t be contributing useful information. Beyond that the columns were renamed from their original questions to a shorter title.

Methodology

To get the data ready for the different algorithms all the categorical responses were one hot encoded. This included questions such as “Did you do the readings”, “Prior Experience”, etc. The quantitative questions remained the same. Beyond this three new columns were made based on the points each student received on the test. The first column being their grade as a number which was simply the score they got divided by the maximum number of points. The second column was their corresponding letter grade based on the following mapping in interval notation:

A: [1 - 0.9] , B: (0.90 - 0.80] , (C: 0.79 - 0.70] , D: (0.69 - 0.60] , F: (0.60 - 0.0]

The third column corresponded to pass or fail where passing was ≥ 0.70 and failing was < 0.70 . With these changes, the data was ready for our machine learning algorithms.

Machine Learning

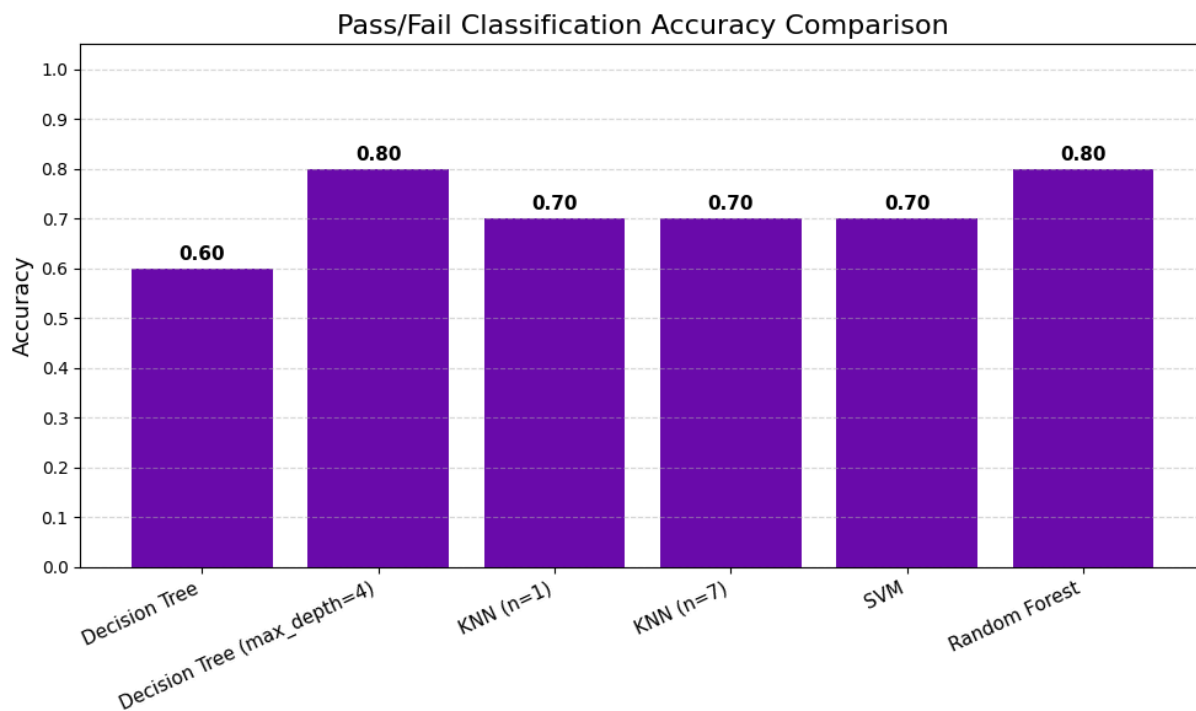
The machine learning testing part of our study was split into three parts.

- Pass/Fail - Classifying based on whether the student got a passing or failing grade
- Letter Grade - Classifying based on whether the student received A,B,C,D, or F
- Grade - Linear Regression based on the student's exact number score.

Pass or Fail

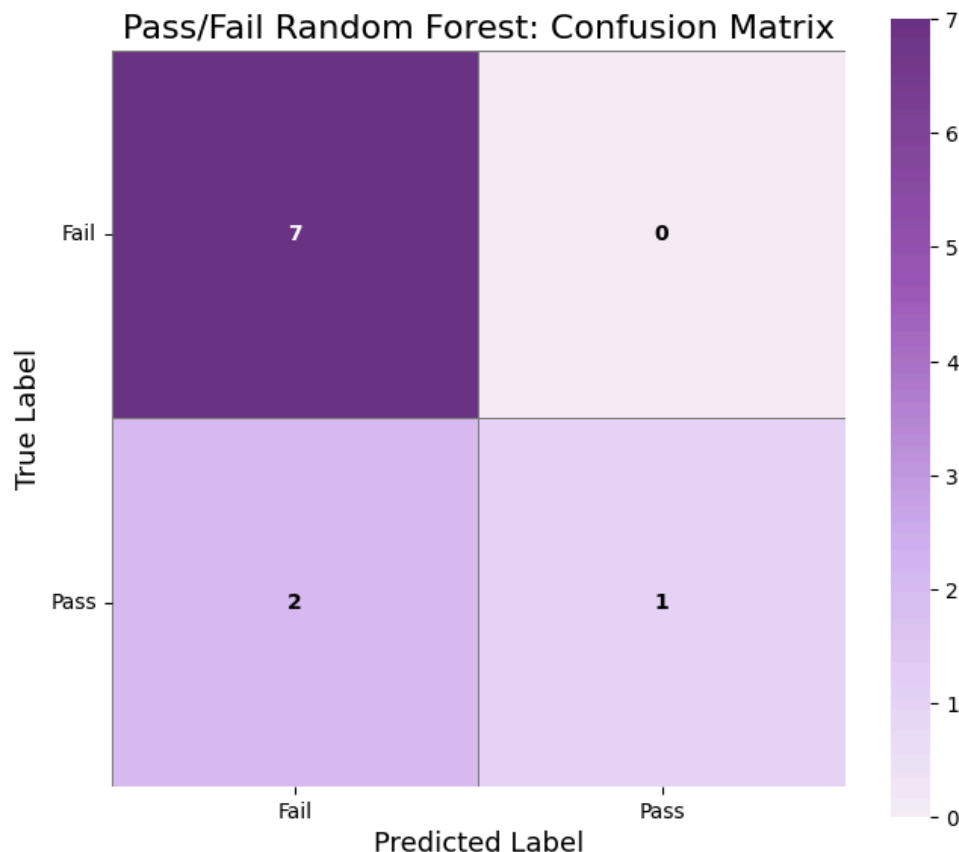
For pass or fail the features categories was set to include all the columns related to the survey which included all the questions as well as the time data, making sure to exclude any data related to the student's score. The label data was our Pass/Fail column. The responses were then split with 90% being used for training and testing done on the last 10%. When it came to hyperparameter values were mostly determined by a guess and check system to get a seemingly optimal value for improved accuracy.

Pass/Fail Classifying	
Algorithm	Accuracy
Decision Tree	0.60
Decision Tree (max_depth = 4)	0.80
KNN (n = 1)	0.70
KNN (n = 7)	0.70
SVM	0.70
Random Forest	0.80



The algorithms used were Decision Tree, KNN, SVM, and Random Forest with decision trees and KNN being ran with different hyperparameters. The resulting accuracies can be summarized in the diagram above. The best accuracy is Random Forest with an accuracy of 0.80 while the lowest is with decision trees with no max depth. This makes sense with our data as the decision tree with no hyper parameter is most likely overfitting during training and during poorly on test data. While the Random Forest is properly weighting the different pieces of data.

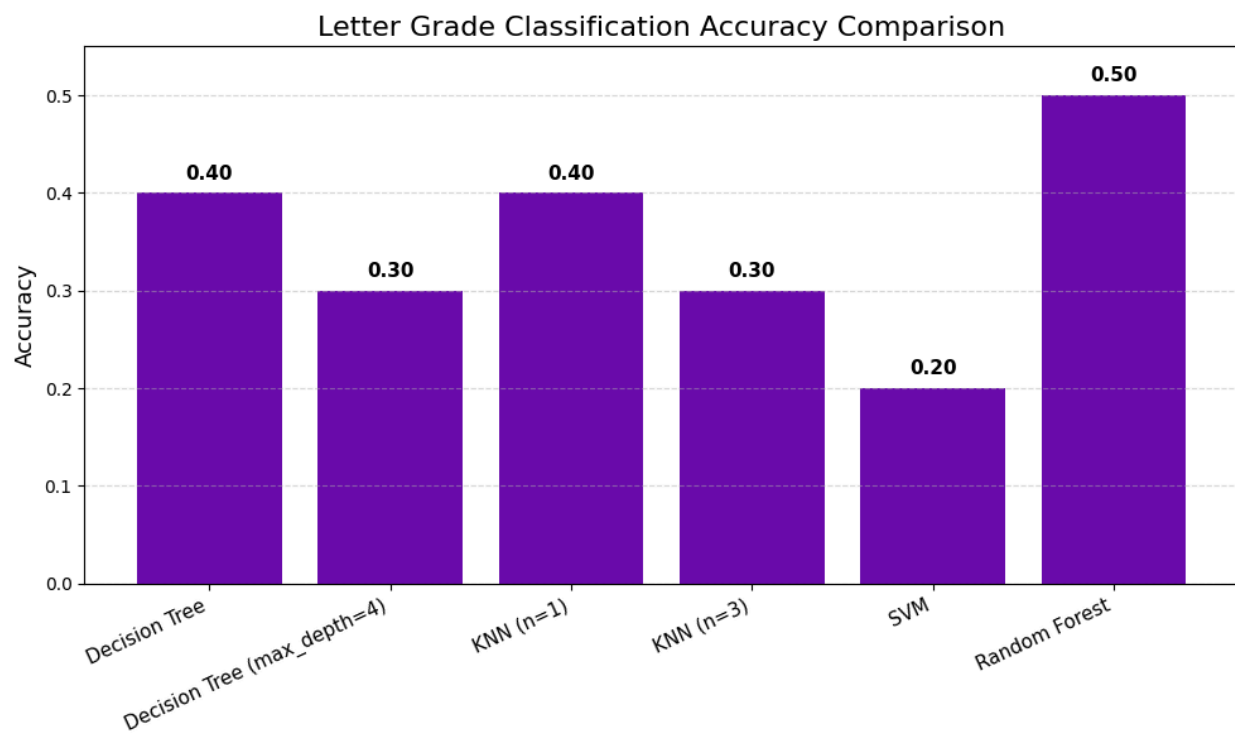
On the surface, these are pretty good accuracies with an especially good accuracy from our decision tree classifications. However, it is important to note that the split of our data was 62 fails with 33 passes. Meaning that our baseline accuracy would be about 65% if all data was labeled as failing. This means that random forest is the only algorithm to be considerably better than the baseline. We can see it's exact results in the confusion matrix to the side.



Letter Grade

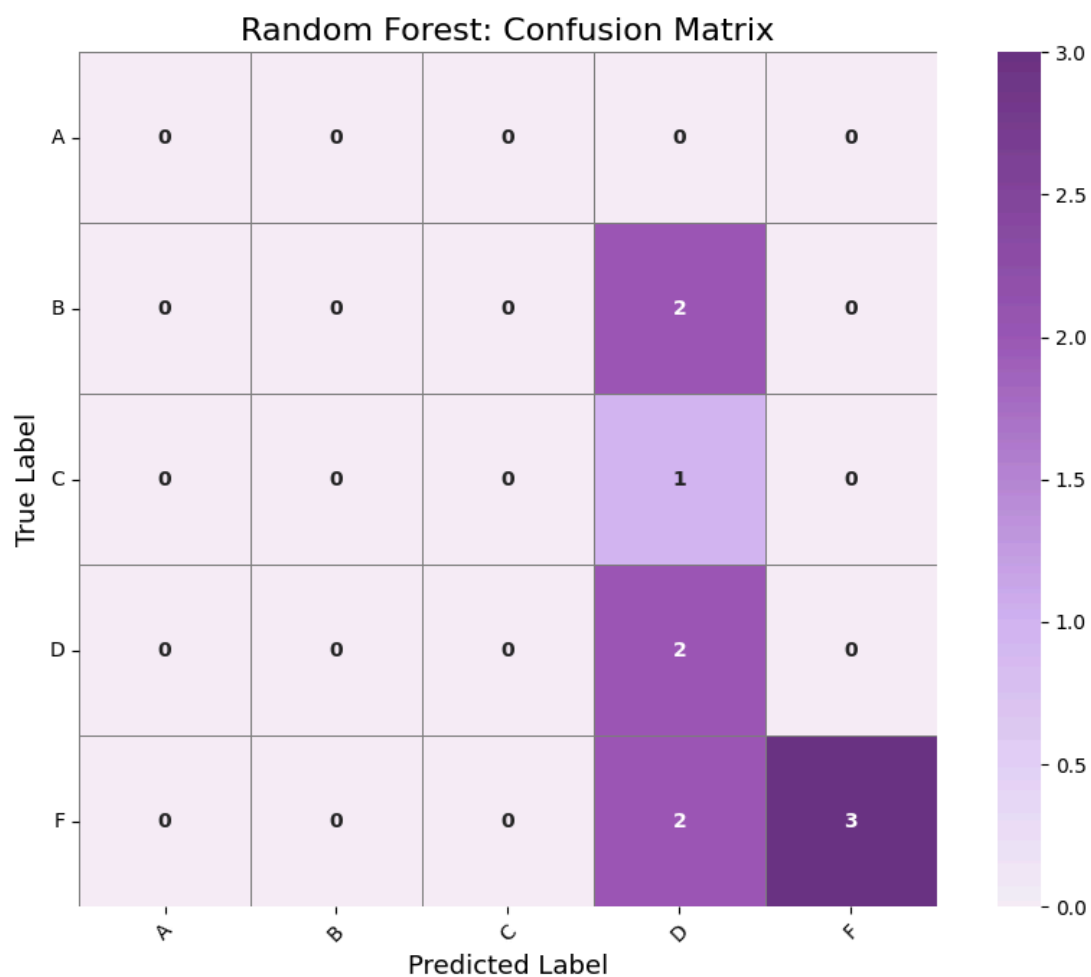
For classifying based on letter grade. Our features set stayed the same consisting of the data from the survey while out label data this time was our letter grade column. We ran the same algorithms using the same 0.1 split.

Letter Grade Classifying	
Algorithm	Accuracy
Decision Tree	0.40
Decision Tree (max_depth = 4)	0.30
KNN (n = 1)	0.40
KNN (n = 3)	0.30
SVM	0.20
Random Forest	0.50



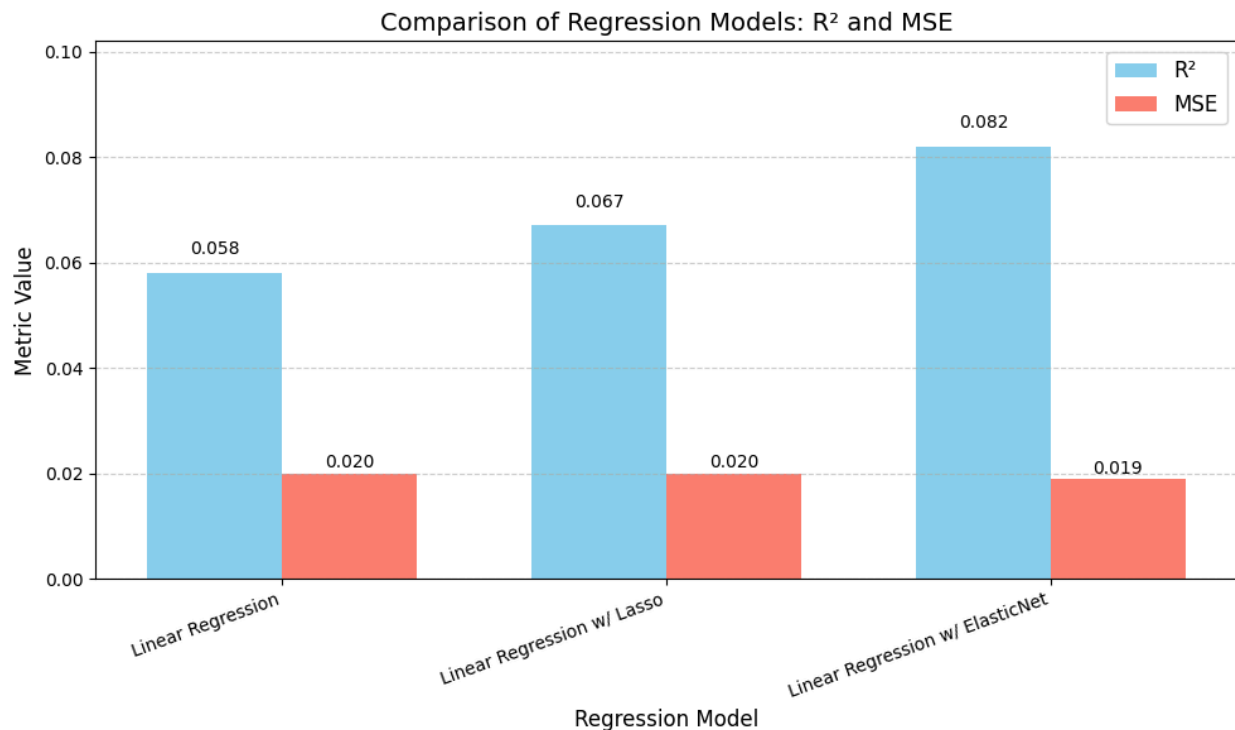
The same algorithms were run with slight tweaking of hyperparameters to better optimize them. The resulting accuracies are noticeably lower with random forest being the highest with an accuracy of 0.50 and our lowest being SVM with an accuracy of 0.20.

Here, our baseline accuracy should be around 0.35 since our split of grades was 1 A, 17 Bs, 15 Cs, 32 Ds, and 35Fs. If we were to label everything as F our accuracy would be sound 0.35. This means are algorithms are not much better than the complete baseline with the only one having any promise being random forests. The exact results can be seen in the following confusion matrix.



Regression

For trying to appoint a specific number grade to each student we can use regression algorithms. Specifically we'll try linear regression followed with Lasso, Ridge, and Elasticnet variations. Our feature set again stays the same looking at the questions from the survey while our label data is our number grade column. Linear regression was then run using the same 90% train 10% test split.



Regression		
Algorithm	R^2	MSE
Linear Regression	0.058	0.020
Linear Regression Lasso	0.067	0.020
*Linear Regression Ridge	0.443	0.020
Linear Regression ElasticNet	0.082	0.019

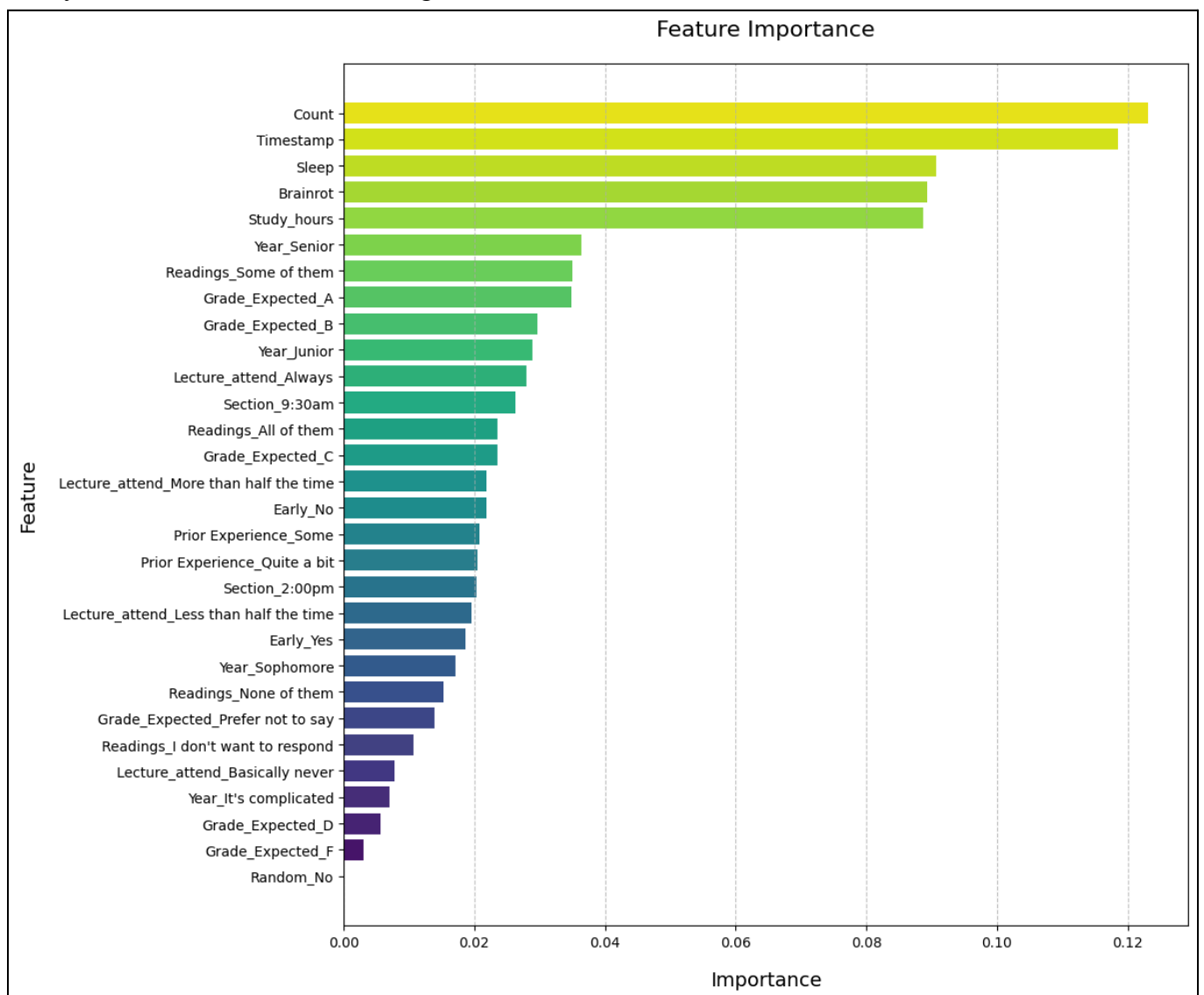
*Linear Regression had a matrix warning, meaning the result may not be accurate.

Our results can be summarized in the above table and visualized in the bar chart further above. Overall regression performed extremely poorly on this data set as a horizontal line with the mean value should be scoring at an r^2 of 0. This means in no proper case was our regression slightly above the base case. This is most likely due to the excessive noise and weird distribution of our data set, as well as the distribution of grades not being linear.

Machine Learning: Biggest Factor

Finally a study was also conducted to see which aspect of our survey was most impactful and predicting student scores. If a meaningful enough result is found this could be used to guide students on how to properly prepare for future exams.

To examine this, a Random Forest was as decisions trees seemed to be our most accurate algorithm. On top of this by looking at which piece of data the trees tend to split at can tell us which data has the most impact on predicting grades. Our features were still the data from the survey and our labels were our letter grade column. Our results were as follows:



Each value corresponds with how important that piece of data was in predicting letter grade through the decision tree. The most interesting part of our results is that the most important piece of information seems to be when the survey was taken whether that be as a time/date or in relation to other students as a count. It is important to not that categorical question like Expected Grade have their importance split up into their different responses so quantitative questions will tend to be larger. However, even factoring that in, Count and Timestamp still lead in importance by a large amount.

This is interesting as it seemingly has nothing to do with the exam or about the students taking the survey. There might be a couple reasons for this. The first is that people who were on top of their things and readily tuned into this class filled out the survey first and these were matches students who were on top of their preparation for the exam. The same goes for the other way around where people who submitted the survey last may have also procrastinated when it came to studying for the exam.

The other idea, is that these were the only two pieces of data that were taken automatically by the survey. They were purely objective and couldn't have been swayed by what the person taking the survey was thinking. For instance, a student could easily overestimate how much studying they did or be ashamed to admit how much time they spent on scrolling apps. This sense of objectivity might be what brought these two pieces of data so far up in importance.

However, our classification algorithm might just be interpreting these pieces of data in a strange way since they are completely different than any of the other columns of information resulting in weird weighting and results.

Ignoring those two it seems that the most reliant factors to look at when determining what influences grades is study hours and scrolling time. These make the most logical sense as studying for the exam should correlate with a higher grade. This is not always the case as realistically students are only studying as much as they need, so a student who studied 1 hour is as prepared as one who studied 8. Scrolling also most likely is correlated with results. I assume it is not a direct causation but you could assume that the more time spent on these apps is less time preparing for the exam.

Conclusion & Future Work

Overall, when classifying our data the best algorithms seem to be decision trees and the worst seems to be forms of regression. This is most likely due to the noise included in our dataset and the wacky distribution of scores.

Using this information however, we were able to examine which factors were most important in predicting exam scores. The most important interestingly being the time at which the survey was taken, with study and social media hours being second and third. This is an interesting correlation as it could be due to the time being automatically recorded by the survey rather than a subjective answer from students. It could also be caused by a correlation between students who submitted the survey early vs those who waited to do and their studying habits. More tests would be needed to explore this further and see if this is just a fluke of our dataset. This could include maybe a hypothesis test on the time the survey was submitted and the result of the exam or a more in depth look at the random forest used to generate these results. Future testing could also use more rigorous techniques for our algorithms like k-fold cross validation or nonlinear regression models which may result in better accuracies for our data. On top of that, our metrics could be used to test our algorithms because accuracy may be inflated by the strange distribution of grades. Something like F1 score might provide more insight (as soon as we figure out what it's used for).